

Math 3790H: Assignment I

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Problem 1. Let A and B be nonempty bounded subsets of $\mathbb{R}$, and let $A + B$ be the set of all sums $a + b$ where $a \in A$ and $b \in B$. Prove that $\sup(A + B) = \sup A + \sup B$.

Hint: To show $\sup A + \sup B \leq \sup(A + B)$ show that for each $b \in B$, $\sup(A + B) - b$ is an upper bound for A , hence $\sup A \leq \sup(A + B) - b$. Then show $\sup(A + B) - \sup A$ is an upper bound for B .

Solution 1. Evidently $\sup A + \sup B \geq \sup(A + B)$ as $\sup A \geq a \forall a \in A$ and $\sup B \geq b \forall b \in B$ and adding these inequalities gives $\sup A + \sup B \geq a + b \forall a \in A, b \in B$ and all $a + b$ form the set $A + B$. Now we need only show the reverse inequality to force equality.

Following the hint, we know that, by definition of the supremum, $a + b \leq \sup(A + B) \forall a + b \in A + B$. Subtracting b from both sides we see that $a \leq \sup(A + B) - b$, hence $\sup(A + B) - b$ forms an upper bound for A and $\sup A \leq \sup(A + B) - b$. Rearranging by adding b and subtracting $\sup A$ we see that $b \leq \sup(A + B) - \sup A$ and hence $\sup(A + B) - \sup A$ forms an upper bound for B , i.e. $\sup B \leq \sup(A + B) - \sup A$. Rearranging by adding $\sup A$ to both sides we obtain the desired inequality $\sup B + \sup A \leq \sup(A + B)$, which forces

$$\sup(A + B) = \sup A + \sup B$$

as we already showed that $\sup A + \sup B \geq \sup(A + B)$.

Problem 2. Let $\mathbb{I}$ be the set of real numbers that are not rational; elements of $\mathbb{I}$ are called irrational numbers. Prove if $a < b$, then there exists $x \in \mathbb{I}$ such that $a < x < b$. Hint: First show $\{r + \sqrt{2} | r \in \mathbb{Q}\} \subseteq \mathbb{I}$.

Solution 2. To get a number anywhere between a and b we take $a + n(b - a)$ for $n < 1$. To use the hint, we take $n = 1/\sqrt{2} = \sqrt{2}/2$, meaning we only need to prove the hint and show we can build $a + (b - a)\sqrt{2}/2$ in the given set. Suppose, by way of contradiction, that $r + \sqrt{2}$ is rational and therefore not in $\mathbb{I}$. This, by the definition of $\mathbb{Q}$, means that $r + \sqrt{2} = m/n$ for integers m, n . Suppose also that $r = p/q$ for $p, q \in \mathbb{Z}$. Then

$$\frac{p}{q} + \sqrt{2} = \frac{m}{n} \implies \sqrt{2} = \frac{m}{n} - \frac{p}{q} \implies \sqrt{2} = \frac{mq - pn}{nq}$$

which would imply $\sqrt{2} \in \mathbb{Q}$, which we know is false, hence $r + \sqrt{2} \notin \mathbb{Q}$ and is then in $\mathbb{I}$ and hence the set of all such things is also included in $\mathbb{I}$. Certainly then $\{sr + s\sqrt{2} | s, r \in \mathbb{Q}\} \subseteq \mathbb{I}$ as multiplying an irrational by a rational still yields an irrational. This allows us to build $a + \sqrt{2}(b - a)/2$ by taking $2a/(b - a) + \sqrt{2}$ and multiplying it by $(b - a)/2$.

Problem 3. Prove that every positive real number has exactly 2 square roots, one positive ($\sqrt{x}$) and one negative ($-\sqrt{x}$).

Solution 3. My original attempt was as follows: Suppose there are two numbers $r_1, r_2 \in \mathbb{R}$ which both square to $a \in \mathbb{R}$, $r_1^2 = a$ and $r_2^2 = a$ and take $r_1 \neq r_2$. Subtracting these two equations we obtain $r_1^2 - r_2^2 = 0$ which we can factor as a difference of squares to obtain $(r_1 - r_2)(r_1 + r_2) = 0$. This implies that $r_1 = r_2$ or $r_1 = -r_2$. We created r_1 and r_2 with the condition that $r_1 \neq r_2$ so then $r_1 = -r_2$. Taking the square root of our original expressions, $r_1 = \sqrt{a}$ and $r_2 = -r_1 = -\sqrt{a} \implies r_1 = -\sqrt{a}$, hence real numbers have both positive and negative roots. To show there are only these two, suppose that there is a third $r_3 \in \mathbb{R}$ with $r_3 \neq r_1$ and $r_3 \neq r_2$ but $r_3^2 = a$, then $r_3^2 - a = 0 \implies (r_3 - \sqrt{a})(r_3 + \sqrt{a}) = 0$ which implies that $r_3 = \sqrt{a}$ or $r_3 = -\sqrt{a}$, but this would imply that either $r_3 = r_1$ or $r_3 = r_2$, so no such third root may exist.

After, I was informed that the desired solution was to do with suprema of sets, so I will also attempt that. Consider the set $S = \{r \in \mathbb{R} | r^2 < a\}$. Since S is nonempty (it contains at least $\min(a, 1)$, since something less than 1 squared is lesser than the number being squared, but the reverse is true for everything above 1 and $1^2 = 1$) and must be bounded above else if $r > \max(a, 1)$ then $r^2 > r > a$ (again by the same shrinking square reasoning, if $a < 1$

then $r^2 < 1$ and hence $r < 1$, but $r > \max(a, 1) = 1$ so contradiction. If $r > \max(a, 1) = a > 1$, then $r > a$ and since $r^2 > r$ for $a > 1$, $r^2 > r > a$, a contradiction, hence no such r may exist) and r^2 cannot be both bigger than and smaller than a . Hence r is an upper bound for S . Since S is both nonempty and bounded, it has a supremum by the completeness axiom (or something under that heading in my notes). Now we want to show that $(\sup S)^2 = a$, which I will do by contradiction.

Suppose first that $(\sup S)^2 < a$. Then by the Archimedean property there exists $n \in \mathbb{N}$ so that $n(a - (\sup S)^2) > 2\sup(S) + 1$ or that

$$a > \frac{2\sup(S) + 1}{n} + (\sup S)^2.$$

Now consider

$$((\sup S) + \frac{1}{n})^2 = (\sup S)^2 + \frac{2\sup(S)}{n} + \frac{1}{n^2}$$

which is bounded above by

$$(\sup S)^2 + \frac{2\sup(S)}{n} + \frac{1}{n}$$

which using the Archimedean property as above is less than a (this was constructed in reverse). Hence, since

$$((\sup S) + \frac{1}{n})^2 < a,$$

$\sup S + 1/n \in S$ and for all n , $\sup S + \frac{1}{n} > \sup S$, a contradiction, meaning our initial assumption that $(\sup S)^2 < a$ is false.

Now suppose that $(\sup S)^2 > a$. We do the same trick by saying that, using the Archimedean principle, $2(\sup S) < ((\sup S)^2 - a)n$ or equivalently

$$a < (\sup S)^2 - \frac{2(\sup S)}{n}.$$

Then consider the expansion of

$$\left(\sup S - \frac{1}{n}\right)^2 = (\sup S)^2 - \frac{2\sup S}{n} + \frac{1}{n^2} > (\sup S)^2 - \frac{2\sup S}{n}$$

which is strictly greater than a by the Archimedean property expression above. But there must be some $r \in S$ that lies between the supremum and the supremum minus a little bit, i.e. $1/n$, because there are no gaps in $\mathbb{R}$ so

$$\sup S - 1/n < r < \sup S$$

and since $(\sup S - 1/n)^2 > a$, squaring gives

$$a < (\sup S - 1/n)^2 < r^2$$

and hence r cannot belong to S as it does not obey the $r^2 < a$ condition, so $(\sup S)^2 > a$ must be false, and hence $(\sup S)^2 = a$ since it is neither greater nor lesser than. Hence there is a number in $\mathbb{R}$ which squares to a , and so a has at least one square root. The reversal to show negative roots I am not so sure about. I think if we reverse the ordering of S by considering $-S$, then $-\sup(S) = \inf(-S)$ as the largest thing in S becomes the smallest in $-S$, with a sign change. Then I *think* that everything we did above holds if you perform that change, with all the orderings reversed of course. That there are exactly these two roots I have no idea how to show using suprema.

Problem 4. Use the $\epsilon - N$ definition of the limit to prove the following

- (a) $\lim_{4n-2} \frac{3n+5}{4n-2} = 3/4$.
- (b) $\lim_{n^5-7} \frac{2n^2+8}{n^5-7} = 0$
- (c) Let (s_n) be a sequence that converges to 2. Show that $\lim(3s_n + 1) = 7$.
- (d) The sequence $(a_n)_{n \in \mathbb{N}}$ where $a_n = \sin(n\pi/2)$, diverges.

Solution 4. (a) We must show that for every $\epsilon > 0$ there is an $N \in \mathbb{N}$ such that for $n > N$,

$$\left| \frac{3n+5}{4n-2} - 3/4 \right| < \epsilon.$$

For the rough work part here we note that at lowest $n = 1$, the expression in absolute value is positive, so we can remove the absolute value,

$$\begin{aligned} &\implies \frac{3n+5}{4n-2} - 3/4 < \epsilon \\ &\Leftrightarrow \frac{12n+20-12n+6}{4n-2} < \epsilon \\ &\Leftrightarrow \frac{13}{2n-1} < \epsilon \\ &\Leftrightarrow \frac{1}{2} \left[\frac{13}{\epsilon} + 1 \right] < n. \end{aligned}$$

Now,

Proof. Consider

$$N = \frac{1}{2} \left[\frac{13}{\epsilon} + 1 \right].$$

Then $n > N$ implies

$$n > \frac{1}{2} \left[\frac{13}{\epsilon} + 1 \right]$$

or

$$2n - 1 > \frac{13}{\epsilon}$$

hence

$$\epsilon > \frac{13}{2n-1}$$

and so (with some foreknowledge from the informal discussion about how we could rewrite this fraction)

$$\epsilon > \frac{3n+5}{4n-2} - 3/4$$

or since both sides are strictly positive

$$\epsilon > \left| \frac{3n+5}{4n-2} - 3/4 \right|.$$

□

(b) We to find n so that

$$\left| \frac{2n^2+8}{n^5-7} - 0 \right| < \epsilon.$$

Here we restrict to $n > 1$ so we can remove the absolute value as $n^5 - 7$ is only positive from there on. To solve for n here it will be easier if we can establish some other S'_n like

$$\frac{2n^2+8}{n^5-7} < S'_n < \epsilon.$$

To do this we establish something that bounds the numerator above (and so is always greater than it, under the condition $n > 1$) and bounds the denominator below (and so is always less than it, making the fraction overall larger). The numerator is bounded above by $5n^2$ for $n > 1$ (to be careful, $n \geq 2$, as their intersection is between 1 and 2, but $n \in \mathbb{N}$) and the denominator is bounded below by n^3 for $n > 1$ hence

$$\frac{2n^2+8}{n^5-7} < \frac{5n^2}{n^3} < \epsilon.$$

Rearranging

$$\frac{5}{\epsilon} < n.$$

Proof. Consider $N = \max\{2, 5/\epsilon\}$ so that $n \geq 2$. $n > N$ implies that

$$n > 5/\epsilon \implies \frac{5}{n} < \epsilon.$$

Certainly then

$$\frac{5n^2}{n^3} < \epsilon$$

and since we force $n > 1$ $n^3 < n^5 < n^5 - 7$ and $5n^2 > 2n^2 + 8$, the first being evident from the degrees of n and the second becoming clear when considering the derivatives of both sides as $10n > 4n$. Hence

$$\frac{2n^2 + 8}{n^5 - 7} < \frac{5n^2}{n^3} < \epsilon$$

and hence

$$\left| \frac{2n^2 + 8}{n^5 - 7} \right| < \epsilon.$$

□

- (c) We seek to show that for all $\epsilon > 0$ we can find an N such that for $n > N$

$$|3s_n + 1 - 7| = |3s_n - 6| = 3|s_n - 2| < \epsilon.$$

Then dividing through by 3, $|s_n - 2| < \epsilon/3$. This is true since it is just the $\epsilon - N$ definition applied to s_n , which we know converges to 2 for any

- (d) Suppose, by way of contradiction, that the limit exists and takes the value L . Then, by definition, there is an $N \in \mathbb{N}$ such that for all $n > N$ and $\epsilon > 0$,

$$\left| \sin\left(\frac{n\pi}{2}\right) - L \right| < \epsilon.$$

$\sin\left(\frac{n\pi}{2}\right)$ takes on three values, 1, 0, and -1 .

Consider the case where we suppose it converges to zero ($L = 0$), Then $\left|\sin\left(\frac{n\pi}{2}\right)\right| < \epsilon$, however for arbitrarily small epsilon an n can always be found so that $\sin\left(\frac{n\pi}{2}\right) = \pm 1$, a contradiction, hence it cannot converge to $L = 0$.

Consider now $L = 1$, then $\left|\sin\left(\frac{n\pi}{2}\right) - 1\right| < \epsilon$. However, as above, an n can always be found so that the term in absolute value comes out to 2 ($\sin\left(\frac{n\pi}{2}\right) = -1$), a contradiction for arbitrarily small ϵ , hence $L \neq 1$.

Consider $L = -1$, then $\left|\sin\left(\frac{n\pi}{2}\right) + 1\right| < \epsilon$. However, as above, an n can always be found so that the term in absolute value comes out to 2 ($\sin\left(\frac{n\pi}{2}\right) = 1$), a contradiction for arbitrarily small ϵ , hence $L \neq -1$. Since the sequence cannot converge to any value that its elements take on, it must diverge.